

Master Colloquium

Prof. Dr. Rand Kouatly

Winter Semester 2026

University of Applied Sciences Europe
Iserlohn · Berlin · Hamburg

Selecting Research Topics

- General Overview
- Recommendation Steps
- Find your supervisors
- Formal Application
- Research Topics

University of Applied Sciences Europe

Prof. Dr. Rand Kouatly

Master Colloquium Module

- This Module is a complement module for your Thesis,
- You will learn:
  - ✓ How you will select the research topic,
  - ✓ Lists of proposed research topic,
  - ✓ Formal Registration and rules,
  - ✓ Thesis rules and regulations,
  - ✓ How you write a professional Thesis and research paper,
  - ✓ How you do a Research and Thesis presentation,

General Overview

- Being able to choose suitable research topics is an important skill to have for any student.
- The importance of writing a good Thesis can lead students to feel an enormous and looming weight hanging over their heads as time passes, however.
- In this lecture, you will see how to find the perfect research topics in just five easy steps, stress-free.

Master Colloquium Module

- Within two weeks or one month from now, Each students has to:
  - ✓ Find a research topic,
  - ✓ Find and agrees with supervisor and co-supervisor,
  - ✓ Prepare a final exam presentation with same topic of his thesis,
  - ✓ The presentation dates are:
  - ✓ **(DS 120 A+B group)**
    - Period 1: 16 Dec 2026 at 14:00 – 16:15,
    - Period 2: 6 Jan 2027 at 14:00 – 16:15,
    - Period 3: 13 Jan 2027 at 14:00 – 16:15,
  - ✓ **(DS 90 group)**
    - Period 1: 16 Dec 2026 at 8:30 – 10:45,
    - Period 2: 6 Jan 2027 at 8:30 – 10:45,
    - Period 3: 13 Jan 2027 at 8:30 – 10:45,
  - ✓ **(SE 60-90-120 group)**
    - Period 1: 16 Dec 2026 at 11:00 – 13:15,
    - Period 2: 6 Jan 2027 at 11:00 – 13:15,
    - Period 3: 13 Jan 2027 at 11:00 – 13:15,

Master Colloquium Module

- ✓ Presentation duration is 10 Minutes with 10 Min Q/A,
- ✓ It is mandatory to all the students attends the exam Period 1, 2 and Period 3, no exception.
- ✓ Presentation 100 points,
- ✓ Not attending the Presentation 0/100 fail the course,
- ✓ Submit a paper in IEEE format, 4 to 8 pages (Optional).

Why Topic Selection Matters

Your thesis is a marathon, not a sprint.

- Motivation: You will spend hundreds of hours on this. Interest fuels persistence.
- Career Trajectory: Shapes your portfolio for PhD applications or industry hiring.
- Mental Health: A well-scoped topic reduces stress; a poorly chosen one leads to burnout.
- The "Make or Break" Point: A focused topic leads to a successful thesis and defense.

How to Begin

- Do not expect choosing a topic for research project to be a quick or easy task!
- You should be thinking about it right from the start of this course (at least right from now).

Phase 1 – Self-Reflection

Start with the "Inside" Work

1. Identify Your Interests
   - Which courses did you enjoy most?
   - What problems do you think about outside of class?

2. Assess Your Skills & Goals
   - Strengths: Leverage what you are good at (data analysis? lab work? writing?, coding, ….).
   - Weaknesses: What skills do you want to develop?

❑ Future Plans:
   - PhD: Choose a topic aligned with potential doctoral advisors.
   - Industry: Choose topics with marketable skills (AI, policy, clinical trials).

Phase 2 – Exploration & Scoping

Ground Ideas in Reality

1. Review the Literature
   - Read recent review articles (Annual Reviews, etc.).
   - Look for sections titled: "Future Directions," "Limitations," "Unanswered Questions.
   - Goal: Find a specific, manageable gap—not a revolution.
2. Consider Your Constraints
   - Time: 10- 16 weeks. Avoid complexed ideas.
   - Resources: Do you need software, or survey? Is it available?
   - Data Access: Is the data actually obtainable? How big and relabels the data?

Phase 3 – The Role of Your Supervisors

They will not do the work on your behave.

1. Find the Right Supervisors (If Not Assigned)
   - Do their research interests overlap with yours?
   - Are they accepting master's students?
   - Do they have capacity to accept additional students?
   - Talk to current students: Is the advisor accessible? Do they help students graduate on time?.

2. Collaborate on Narrowing
   - Come with options, not just a vague idea.
   - Ask: "I'm interested in X. I've read Papers A and B. I'm considering either Focus 1 or Focus 2. Which is more feasible given our resources?"

Phase 4 – Evaluation Criteria

A strong master's topic should be:

1. Feasible: Can I finish this in 10 weeks with the resources you have?
2. Original (enough): Is there a twist (new context, method, or population) on existing research?
3. Significant (to you): Will I still care about this at 3:00 AM during writing?
4. Aligned: Does my supervisor have expertise in this specific area?, With you field of study?
5. Scalable If Plan A fails (e.g., no survey responses), is there a Plan B?

How to approach your chosen topic

Step 1: Understand what the topic is about.

- What are the key words in the question/topic?
- Which questions am I going to answer?
- Which sphere of knowledge does it cover?
- What variables will be considered?
- Etc. etc.

How to approach your chosen topic

Step 2: Review related literature to help refine how you will approach the topic and finding a way to analyze it.

- Focus on identifying sources (literature) that support as well as refute this position
  - Has anybody written about this?
  - If yes, what is it about specifically?
  - If no, what gap is my study filling in?
  - Search for respective and cited research entities (IEEE, Springer, …)

How to approach your chosen topic

CAUTION:

- It is ok to review related literature to help refine how you will approach analyzing a topic, and then discover that the topic is not all that interesting after all.
- In that case, you can choose another topic.
- Do not wait **too long** to make a switch, and
- Be sure to consult with your supervisor first.

How to approach your chosen topic

where, when & how:

- Usually, students ask
  “How do I choose a topic?”

- If you lack ideas, or wish to gain focus, PERFORM some or all of the following ACTIVITIES/strategies:

How to approach your chosen topic

Activity 1: Courses/ readings:

- Review your course readings, particularly the suggested readings, for topic ideas.

- Do not just review what you have already read but jump ahead in the syllabus to readings that have not been covered yet.

How to approach your chosen topic

Activity 2: Libraries

- Search the library catalog for a good, recent introductory books and, if appropriate, more specialized works related to the discipline/area of your course

How to approach your chosen topic

Activity 3
Browse through some current journals in your subject discipline

- Even if most of the articles are not relevant, you can skim through the contents quickly.
- You only need one to be the spark that begins the process of wanting to learn more about a topic.
- Consult a librarian or your supervisor about the core journals within your subject discipline.

How to approach your chosen topic

Activity 4: Past essays

- Think about essays projects you have written for past classes and other coursework you have taken or academic lectures and programs you have attended.
- Thinking back, what most interested you?
- What would you like to know more about?

How to approach your chosen topic

Activity 5: Online portals

- Search online media sources, such as IEEE spectrum magazine, etc., to see if your idea has been covered in the news

- Use this coverage to refine your idea into something that you would like to investigate further but in a more deliberate, scholarly way based on a problem to research

How to approach your chosen topic

Activity 6: Surroundings

- More commonly, look around your workshop, department, university environment or society at large to identify issues of concern to you

Considerations For Selecting A Research Problem

1. Interest

- A research endeavour is usually time consuming, and involves hard work and possibly unforeseen problems.
- One should select a topic of great interest to sustain the required motivation.

Considerations For Selecting A Research Problem

2. Magnitude (scope)

- It is extremely important to select a topic that you can manage within the time and resources at your disposal.

- Narrow the topic down to something manageable, specific and clear.

Considerations For Selecting A Research Problem

3. Measurement Of Concepts

- Make sure that you are clear about the indicators and measurement of concepts (if used) in your study.
- e.g.

How do you measure these words?
“congestion”, “efficiency”, “growth”, “malfunction”, etc.

Considerations For Selecting A Research Problem

4. Level Of Expertise

- Make sure that you have adequate knowledge or expertise for the task you are proposing since you need to do the work yourself.

Considerations For Selecting A Research Problem

5. Relevance/Significance

- Ensure that your study
  1. adds to the existing body of knowledge,
  2. bridges current gaps in knowledge, and
  3. is useful in policy formulation.

- This will help you to sustain interest in the study.

Considerations For Selecting A Research Problem

6. Availability Of Data

- Before finalizing on your topic, make sure that data are available.
- Getting information for your research is very critical to any research.

Considerations For Selecting A Research Problem

7. Ethical Issues

- Ethical issues are research processes and findings that affect the sensibilities and rights of research participants.

- The effect the study on the population and how these ethical problems can be overcome should be thoroughly examined at the problem formulating stage.

Formal requirements

Deadline

- **Is non-negotiable!**
- Ask for the extension in a timely manner – a week before the thesis is due
- If you fall sick in between, get a doctor's note and sent it to the examination office!
- Do not let the deadline pass and then tell everybody about the misfortune that ha befallen you!

Formal requirements

Supervisors

- You need two supervisors
  - One of the two has to be a tenured professor of this university – Prof. Dr.!
- The first supervisor will do the supervising and mentoring
- The second supervisor does only read the thesis
- External lecturers will usually only be first supervisor as second supervisors are not being paid for their efforts
- Ask your supervisor for feedback **before** your hand in the thesis
  - Once you submit it to the examination office, it is too late

Thesis Proposal

Thesis Proposal Structure

1. Title: Concise and specific Reflects the main topic and variables
   - Example: “Design and Implementation of a File-Based Dental Appointment Reservation System in Java”
2. Introduction / Background
   - Context of the study
   - What is already known
   - What problem exists
   - Tip: Start broad, then narrow down to your specific issue.

Thesis Proposal

Contents

3. Problem Statement

- Clearly define the issue your research addresses
- Should be specific and researchable
- Example:
  - Manual appointment systems are inefficient
  - Lack of database-free solutions for small clinics

4. Objectives

- What you aim to achieve
- General Objective: Main goal of the study
- Specific Objectives: Break down into smaller steps
- Example:
  - To design a Java-based appointment system
  - To implement file-based data storageTo evaluate system

Thesis Proposal

Contents

5. Research Questions

- Questions your study will answer
  - Example: How effective is a file-based system compared to database systems?
  - Can the system handle multiple bookings efficiently

6. Significance of the Study

- Why your research matters
- Who benefits (e.g., clinics, developers, students)

7. Scope and Limitations

- What is included and excluded
- Example:
  - Includes appointment scheduling
  - Excludes online payment integration

Thesis Proposal

Contents

8. Methodology
- How you will conduct the research
9. Expected Outcomes
- What you expect to achieve
- Example: A functional desktop system
    - Improved scheduling efficiency
10. Timeline
- Project schedule (Gantt chart or phases)
11. References
- List of sources (APA, IEEE, etc.)

Formal requirements

Recommended Supervisors,

- First Supervisor:
  - Prof. Dr. Rand Kouatly, rand.kouatly@ue-germany.de
  - Prof. Dr. Talha Ali Khan, talhaali.khan@ue-germany.de
  - Prof. Dr. Iftikhar Ahmed, iftikhar.ahmed@ue-germany.de
  - Prof. Dr. Raja Hashim Ali, hashim.ali@ue-germany.de
  - Prof. Dr. Aaron Crandall, Aaron.Crandall@ue-germany.de
  - Prof. Dr. Ahmed Abdeltawab, ahmed.abdeltawab@ue-germany.de
  - Prof. Dr. Souad El Hassanie, souad.elhassanie@ue-germany.de
  - Dr. Farhan Khan, farhan.khan@ue-germany.de
- Second Supervisors:
  - Dr. Ila Chandrakar, ila.chandrakar@ue-germany.de
  - Dr. Harsha Raju, harsha.raju@ue-germany.de
  - Dr. Shan Faiz, Shan.Faiz@ue-germany.de

Formal requirements

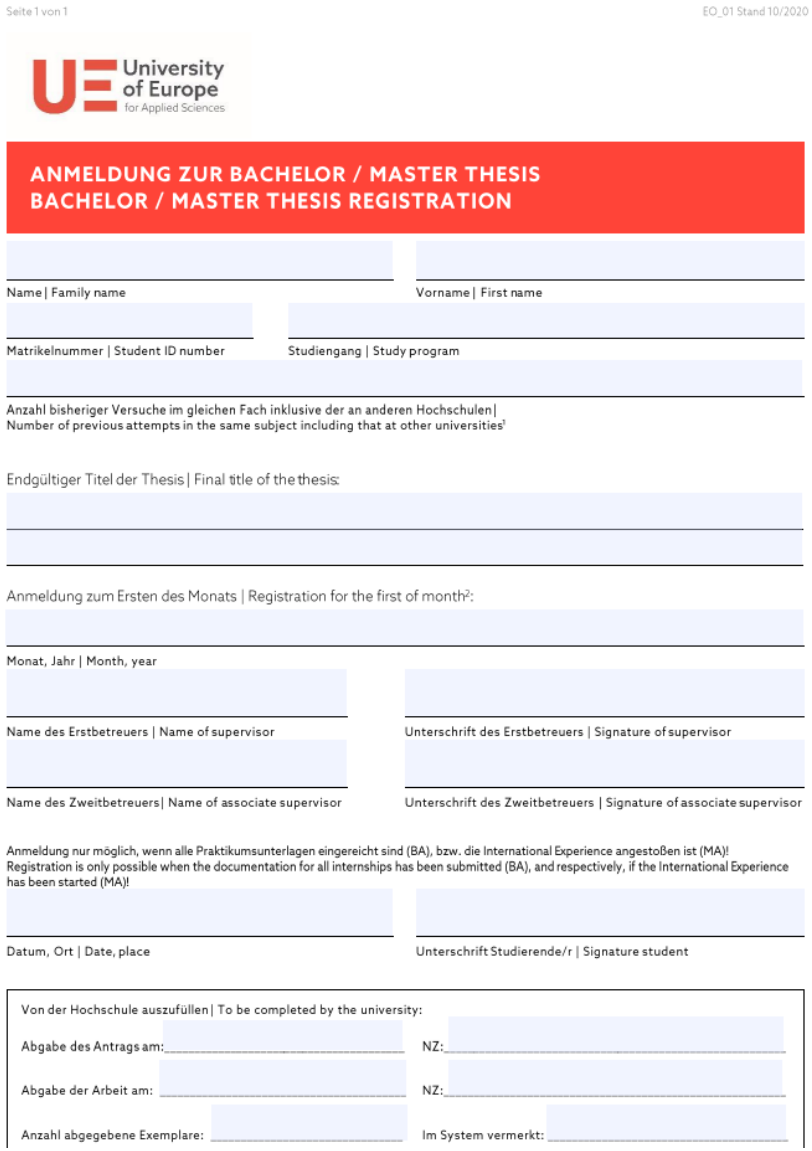
Failed Attempt

- There are many reasons to fail your first attempt
    1. Plagiarism,
    2. Using AI written Thesis,
    3. Missing the deadline,
    4. Writing an essay but not a thesis,
    5. Writing on a level not appropriate for a master of science degree,
    6. Missing the minimum requirements concerning signs,
- You have a second attempt but not a third!
- You have to re-register under a new topic
- You loose time and money. Better get it right the first time!

Formal requirements

Official Registration

- You need to feel in the following form
- Signed by both Supervisors
- Submitted to Students service
- Before submission:
  - ✓ Be sure about the Thesis title, You cannot changed,
  - ✓ Be sure about the registration date,



Seite 1 von 1 EO_01 Stand 10/2020

University of Europe for Applied Sciences

ANMELDUNG ZUR BACHELOR / MASTER THESIS
BACHELOR / MASTER THESIS REGISTRATION

Name | Family name

Vorname | First name

Matrikelnummer | Student ID number

Studiengang | Study program

Anzahl bisheriger Versuche im gleichen Fach inklusive der an anderen Hochschulen |
Number of previous attempts in the same subject including that at other universities[1]

Endgültiger Titel der Thesis | Final title of the thesis:

Anmeldung zum Ersten des Monats | Registration for the first of month[2]:

Monat, Jahr | Month, year

Name des Erstbetreuers | Name of supervisor

Unterschrift des Erstbetreuers | Signature of supervisor

Name des Zweitbetreuers | Name of associate supervisor

Unterschrift des Zweitbetreuers | Signature of associate supervisor

Anmeldung nur möglich, wenn alle Praktikumsunterlagen eingereicht sind (BA), bzw. die International Experience angestoßen ist (MA)!
Registration is only possible when the documentation for all internships has been submitted (BA), and respectively, if the International Experience has been started (MA)!

Datum, Ort | Date, place

Unterschrift Studierende/r | Signature student

Von der Hochschule auszufüllen | To be completed by the university:

Abgabe des Antrags am: NZ:

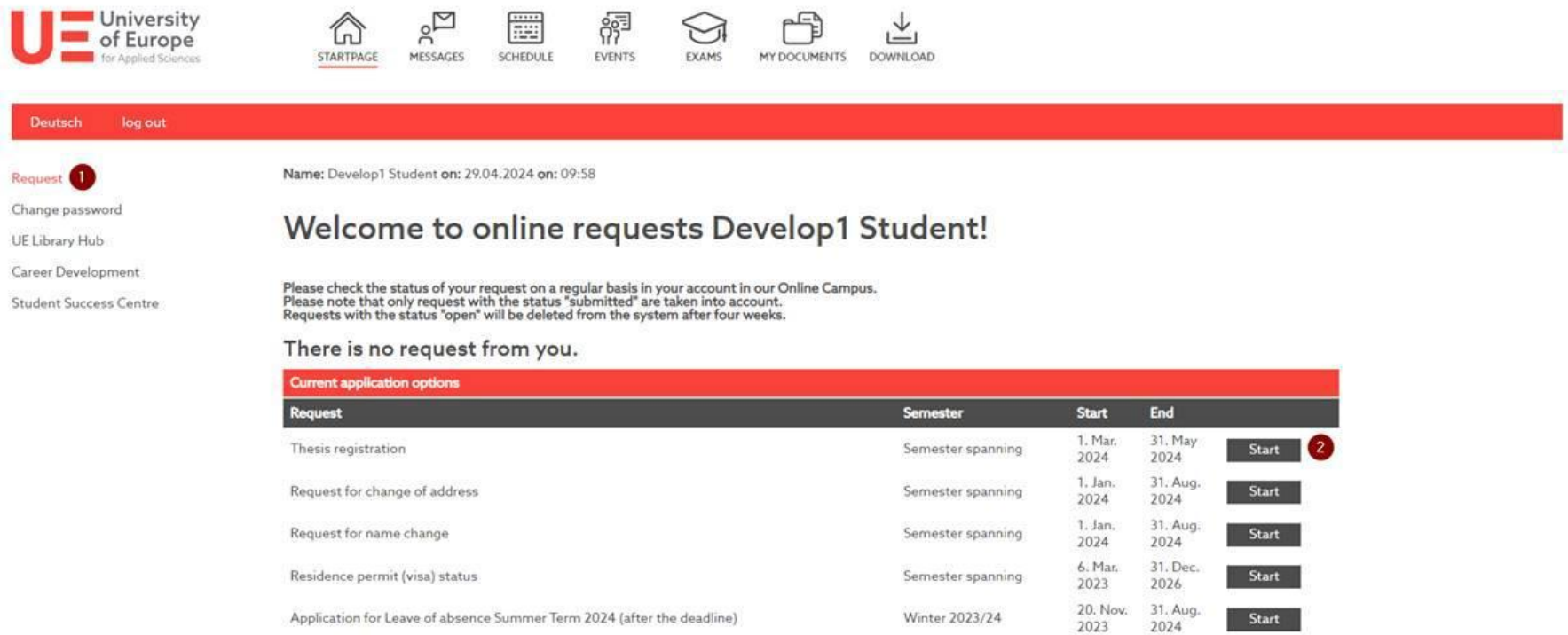
Abgabe der Arbeit am: NZ:

Anzahl abgegebene Exemplare: Im System vermerkt:

Formal requirements

Online Registration

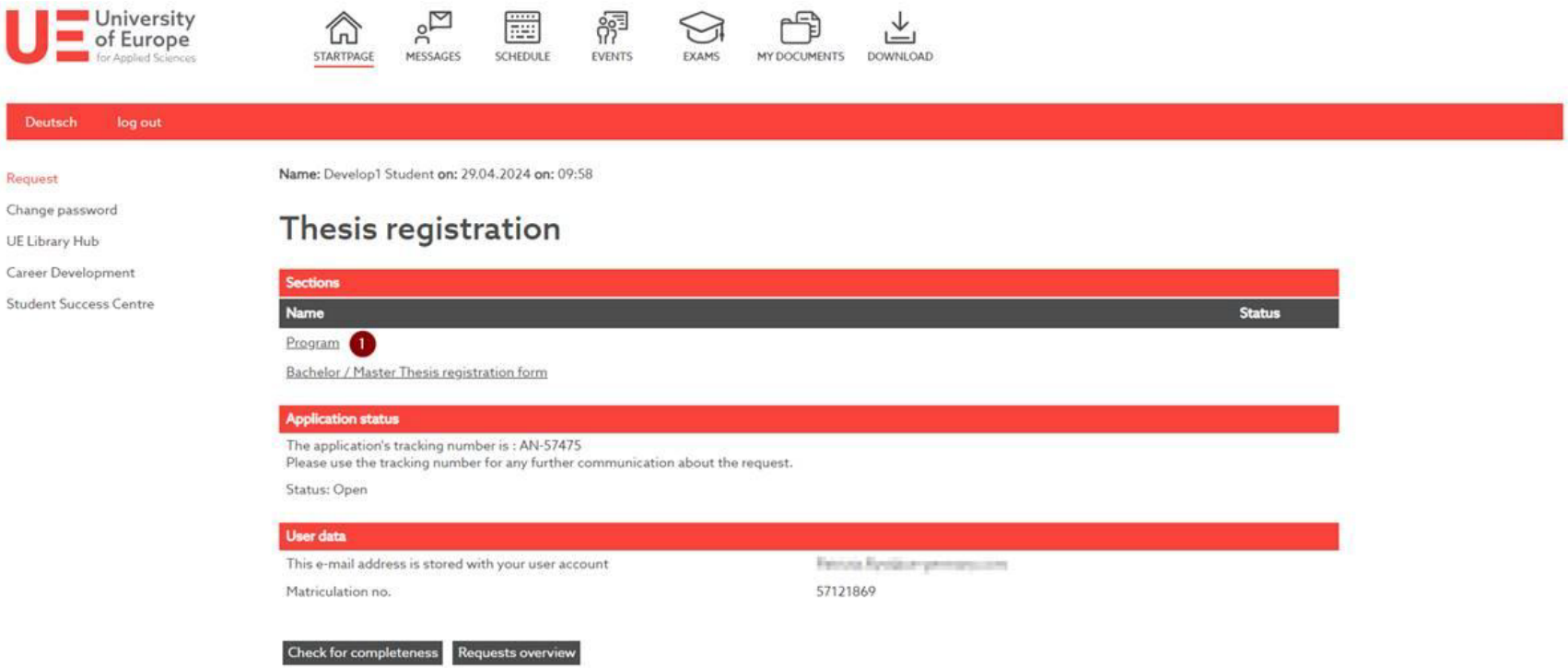
- You need to register online from online CampusNet



Formal requirements

Online Registration

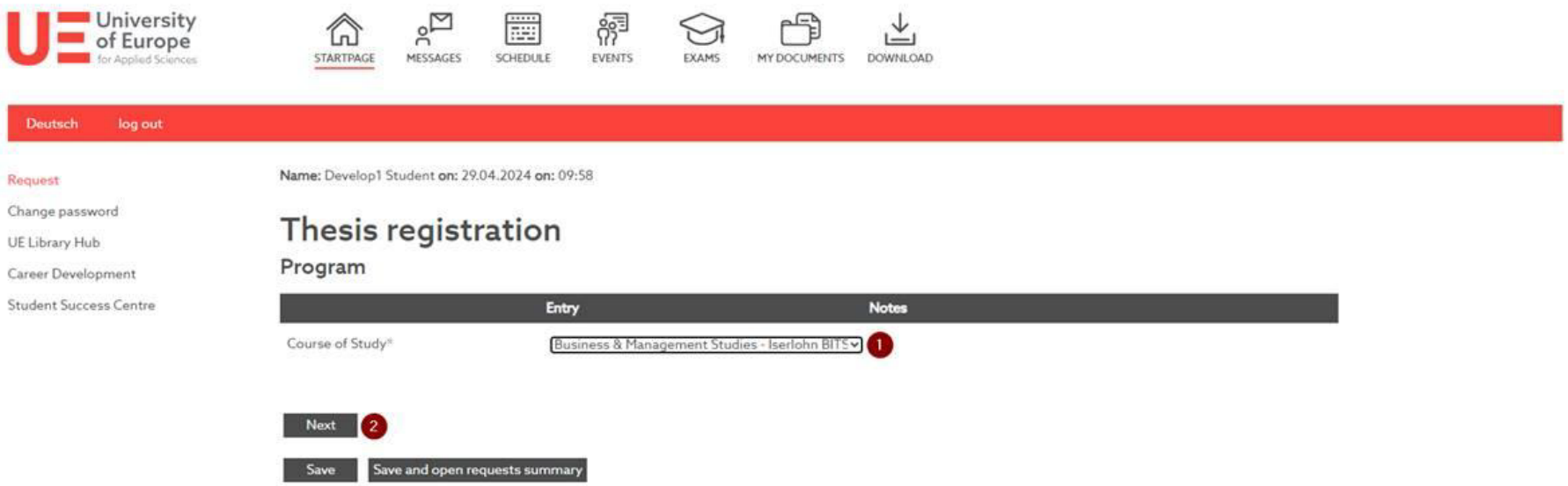
- Step 3: Select the program



Formal requirements

Online Registration

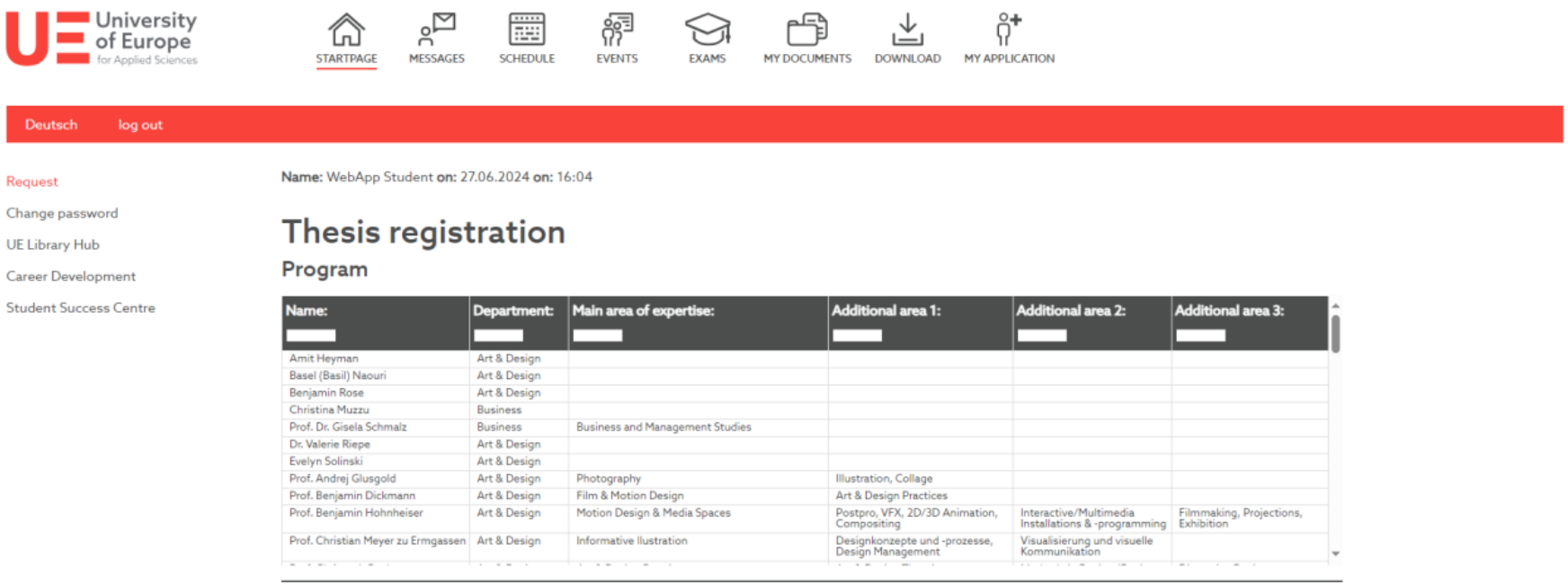
- Step 4: Select the program
- Click “Save” and “Next”



Formal requirements

Online Registration

- Step 5: Check available supervisor and their area of expertise:
  - Choose “Rand Kouatly" as Chaired Supervisor
  - Type of Supervisor "1st Supervisor"Choose "XXXXXXXXXX" in the "Supervisor" field.



Formal requirements

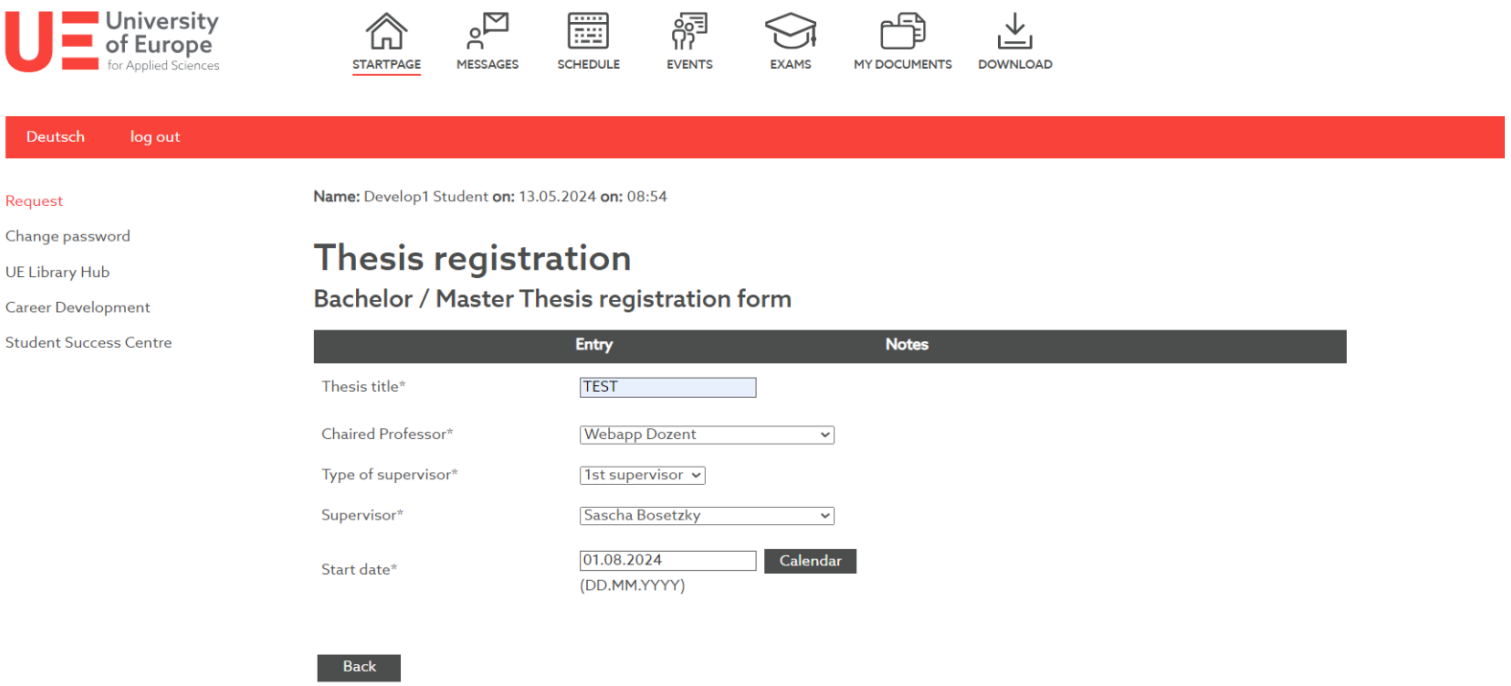
Online Registration

- Step 6: Enter:
  - Thesis title,
  - For example choose “Rand Kouatly" as Chaired Supervisor
  - Type of Supervisor "1st Supervisor"Choose "XXXXXXXXXX" in the "Supervisor" field.
  - Thesis start date,
  - Click “Save and open request summary”

Formal requirements

Online Registration

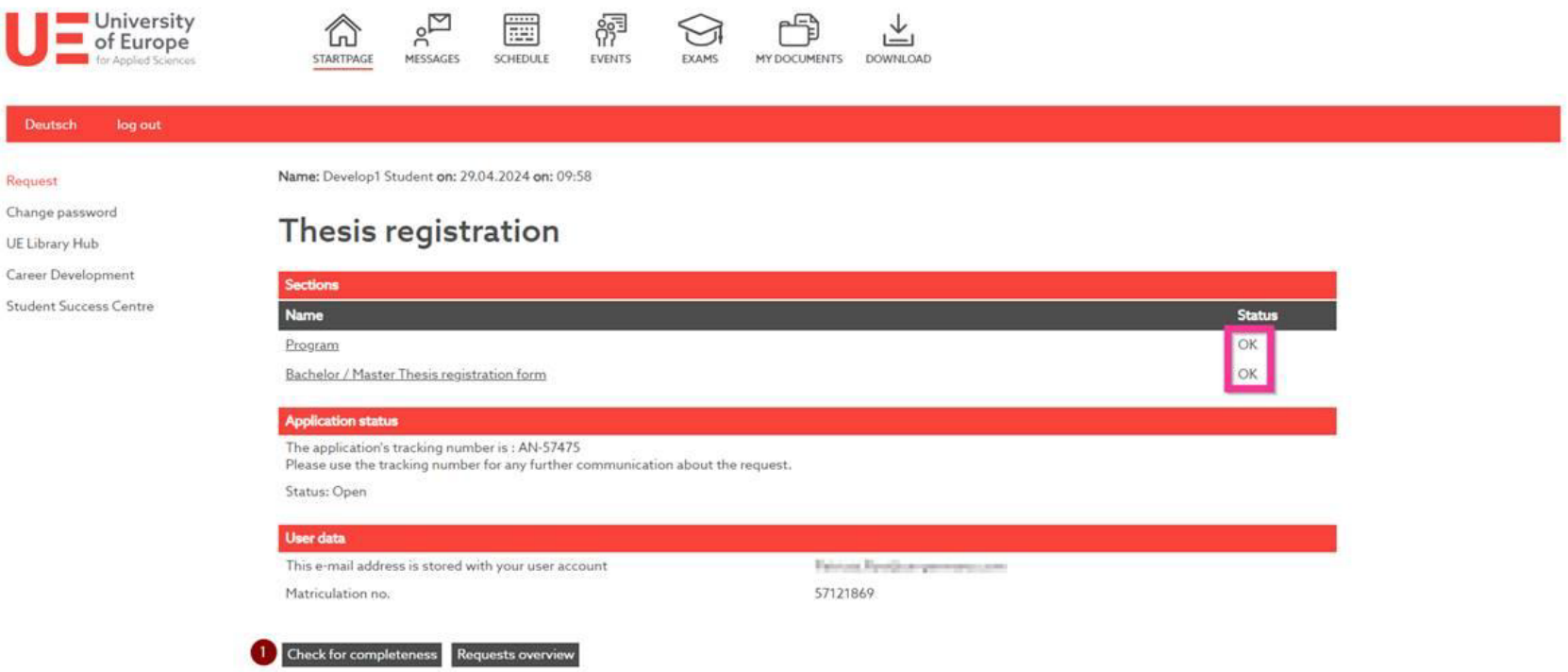
- Step 6: Enter:



Formal requirements

Online Registration

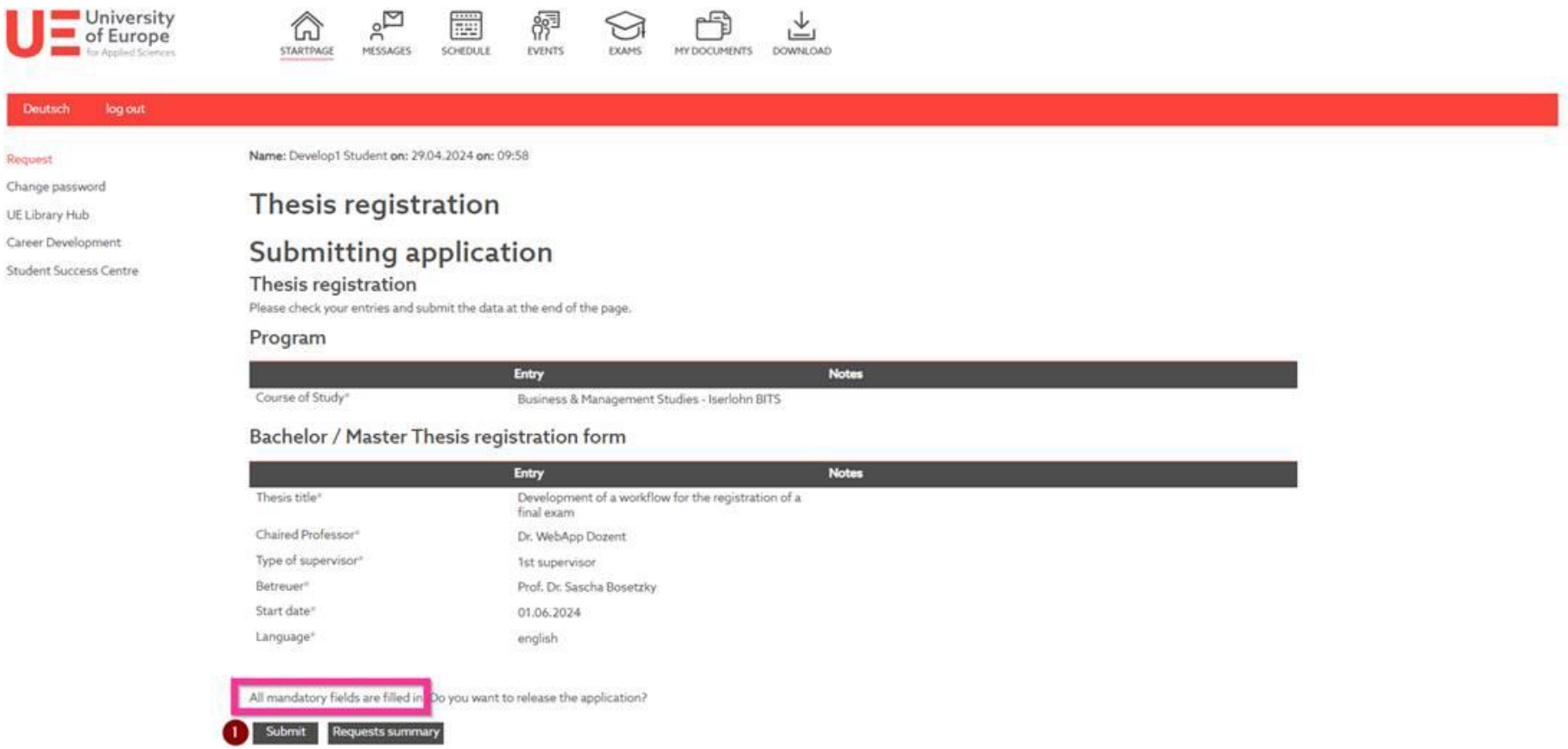
- Step 7: Check if Sections “Program” and “Bachelor / Master Thesis registration form” is OK.



Formal requirements

Online Registration

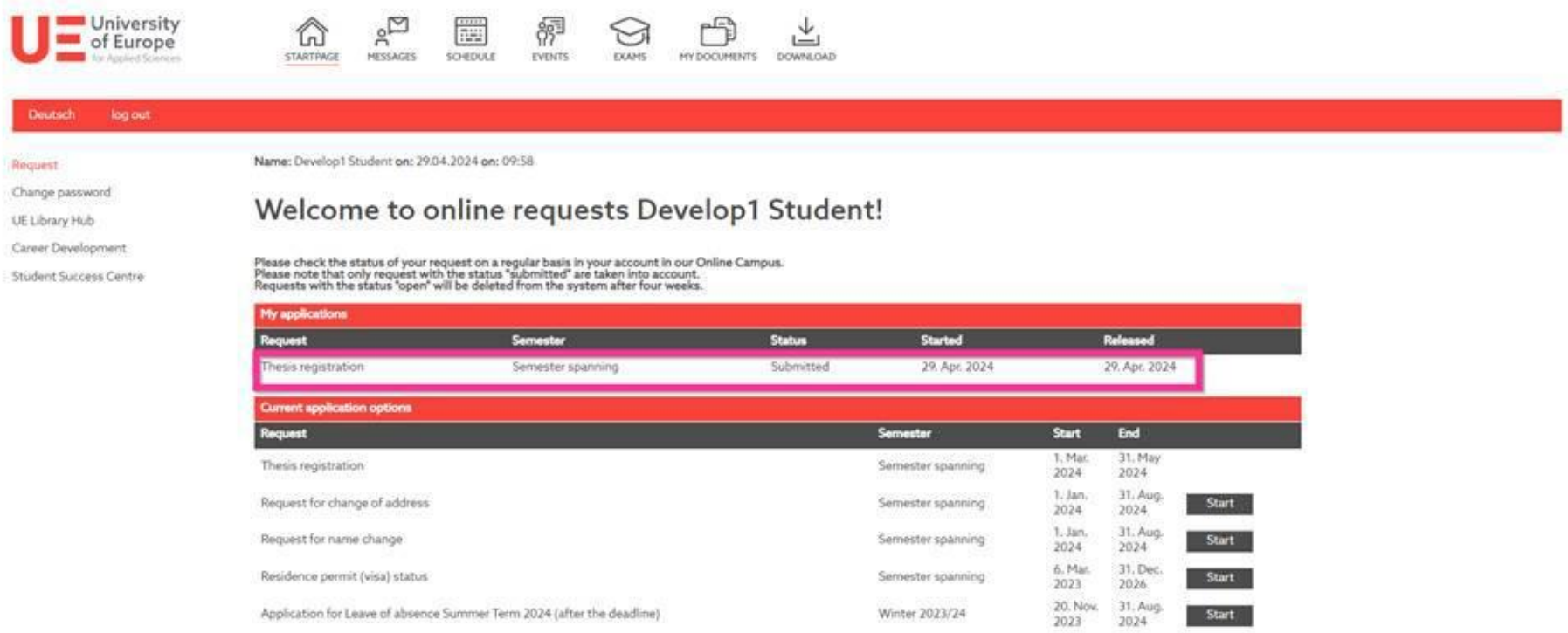
- Step 8: Check all entered data here. If complete and correct click “Submit”.



Formal requirements

Online Registration

- Step 9: Status is now is “Submitted”.



Formal requirements

Thesis Extension

- You need to feel in the following form
- Signed by first Supervisor
- Submitted to Students service before the submission date 10 days.
- Maximum 21 days, 3 weeks

University of Applied Sciences Europe
Iserlohn · Berlin · Hamburg

BA/MA THESIS WORKING PERIOD EXTENSION

1. Applicant

Name _____ Last Name _____

Matriculation number _____ Study program _____

I hereby request a_____ day/week extension of the thesis working period.*

Type of thesis

☐ Bachelor-Thesis ☐ Master-Thesis

☐ Bachelor-Project ☐ Master-Project

Reason for request

☐ Sickness
medical certificate attached

☐ Other
with justification and opinion of the examiner; if applicable, evidence attached

Reason:

_____

_____

_____

_____

Place, date _____ Signature _____

* Please delete where not applicable.

2. Examiner's statement (not applicable in case of illness)

Formal requirements

Failed Attempt

- There are many reasons to fail your first attempt
    1. Plagiarism
    2. Missing the deadline
    3. Writing an essay but not a thesis
    4. Writing on a level not appropriate for a bachelor/master of science degree
    5. Missing the minimum requirements concerning signs
- You have a second attempt but not a third!
- You have to re-register under a new topic
- You loose time and money. Better get it right the first time!

Formal requirements

Attention to detail

- Please read the email carefully that the examination sends you to confirm your registration
- Please read the “Handreichung zum wissenschaftlichen Arbeiten” brochure carefully
- Get feedback from your supervisor!

Research Topics

- **1- Continuous Experimentation:**

Software product and service companies need capabilities to evaluate their development decisions and customer and user value.

Continuous experimentation, as an experiment-driven development approach, may reduce such development risks by iteratively testing product and service assumptions that are critical to the success of the software.

Experiment-driven development has been a crucial component of software development in especially in last decade, companies such as Microsoft, Facebook, Google, Amazon and many others often conduct experiments to base their development decisions on data collected from field usage.

https://researchportal.helsinki.fi/en/publications/the-right-model-for-continuous-experimentation

Research Topics

- **2- Digitalization and digital transformations: impacts on software engineering and systems development:**

Digitalization is nowdays cross-cutting and inherent in most areas of businesses and organizations.

Software is increasingly built-in and ubiquitous. Such trends and developments bring up many potential software research problems, such as:

  - What does digitalization actually entail in different contexts?
  - How should digitalization be taken into account in software development processes?

Research Topics

- **2- Digitalization and digital transformations: impacts on software engineering and systems development:**
  - What is the role of customer/user involvement in software-intensive systems development (e.g., digital services)?
  - What are the key quality attributes?
  - What new software engineering skills and competencies may be needed?
  - What is the role of software (and IT) in general in different digital transformations (e.g., vs. business process development)?
  - How is digitalization related to traditional software engineering and computer science disciplines in different contexts?
  - What aspects of software development and digital technologies are fundamentally new or different from the past?

Research Topics

- **3- Software innovation:**

How is innovation and creativity taken into account in software development processes and methods (e.g., Agile)?

What is the role of customer/user input and feedback in software(-intensive) product creation (e.g., open innovation)?

How to define and measure 'innovativeness' in software development? What makes software development organizations (more) innovative?

https://www.cs.helsinki.fi/en/node/65142

Research Topics

- **4- Software Module Clustering:**

Software module clustering is an unsupervised learning method used to cluster software entities (e.g., classes, modules, or files) of similar features.

The obtained clusters may be used to study, analyze, and understand the structure and behavior of the software entities. Implementing software module clustering with optimal results is challenging.

Several research gaps and challenges in software module clustering are discussed in this paper to provide a useful reference for researchers in this field.

https://ieeexplore.ieee.org/abstract/document/9282200

Research Topics

- **5- Based Software Fault Localization:**

In software debugging, fault localization is the most difficult, expensive, tedious, and time-consuming task, particularly for large-scale software systems.

This is due to the fact that it requires significant human participation and it is difficult to automate its sub-tasks.

Therefore, there is a high demand for automatic fault localization techniques that can help software engineers effectively find the locations of faults with minimal human intervention.

https://ieeexplore.ieee.org/abstract/document/9684433

Research Topics

- **6- Use of Augmented Analytics:**

What is augmented analytics? AA is a concept of data analytics that uses AI, machine learning, and natural language processing to automate the analysis of massive data. What is normally handled by a data scientist is now being automated in delivering insights in real-time.

It takes less time for enterprises to process the data and derives insights from it. The result is also more accurate, thus leading to better decisions. From assisting with data preparation to data processing, analytics, and visualization, AI, ML, and NLP help experts explore data and generate in-depth reports and predictions. Data from within the enterprise and outside the enterprise can be combined through augmented analytics.

Research Topics

- **7- Use of Augmented Analytics:**

The automation of cloud computing services for public and private clouds is achieved using artificial intelligence and machine learning. AIOps is artificial intelligence for IT operations. This is bringing a change in the way enterprises look at big data and cloud services by offering more data security, scalability, centralized database and governance system, and ownership of data at low cost.

One of the big data predictions for 2022 is the increase in the use of hybrid cloud services. A hybrid cloud is an amalgamation of a public cloud and a private cloud platform.

Public clouds are cost-effective but do not provide high data security. A private cloud is more secure but expensive and not a feasible option for all SMEs. The feasible solution is a combination of both where cost and security are balanced to offer more agility. A hybrid cloud helps optimize the resources and performance of the enterprise.

Research Topics

- **8- Hyperautomation:**

Another dominant trend in data science in 2022 is hyper-automation, which began in 2020. Brian Burke, Research Vice President of Gartner, has once said that hyper-automation is inevitable and irreversible, and anything and everything that can be automated should be automated to improve efficiency.

By combining automation with artificial intelligence, machine learning, and smart business processes, you can unlock a higher level of digital transformation in your enterprise. Advanced analytics, business process management, and robotic process automation are considered the core concepts of hyper-automation. The trend is all set to grow in the next few years, with more emphasis on robotic process automation (RPA).

Research Topics

- **9- Automation of Data Cleaning:**

For advanced analytics in 2022, having data is not sufficient. It also refers to incorrect data, data redundancy, and duplicate data with no structure or format.

This causes the data retrieval process to slow down. That directly leads to the loss of time and money for enterprises. On a large scale, this loss could be counted in millions. Many researchers and enterprises are looking for ways to automate data cleaning or scrubbing to speed up data analytics and gain accurate insights from big data. Artificial intelligence and machine learning will play a major role in data cleaning automation.

Research Topics

- **10- Quantum Computing for Faster Analysis :**

One of the trending research topics in data science is Quantum computing. Google is already working on this, where decisions are not taken by the binary digits 0 and 1. The decisions are made using quantum bits of a processor called Sycamore. This processor is said to solve a problem in just 200 seconds.

However, Quantum computing is very much in its early stages and needs a lot of fine-tuning before it can be adopted by a range of enterprises in different industries. Nevertheless, it has started to make its presence felt and will soon become an integral part of business processes. The aim of using Quantum computing is to integrate data by comparing data sets for faster analysis. It also helps in understanding the relationship between two or more models.


Thanks for listening and good luck with the examin!
University of Applied Sciences Europe
Iserlohn · Berlin · Hamburg